
SEMRING TENSOR NETWORKS

RESEARCH NOTES IN THE ENEXA AND QROM PROJECTS

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See Liu et al. (2023, 2021).

1 Semiring

Definition 1. A semiring is a tuple $(S, \oplus, \otimes, 0, 1)$, where S is a set, $0, 1 \in S$, and

$$\oplus : S \times S \rightarrow S \quad , \quad \otimes : S \times S \rightarrow S ,$$

such that the following properties hold:

(i) $(S, \oplus, 0)$ is a commutative monoid¹.

(ii) $(S, \otimes, 1)$ is a commutative monoid.

(iii) $\otimes$ distributes over $\oplus$, that is for any $s_0, s_1, s_2 \in S$

$$s_0 \otimes (s_1 \oplus s_2) = (s_0 \otimes s_1) \oplus (s_0 \otimes s_2) .$$

(iv) 0 annihilates $\otimes$, that is for any $s_0 \in S$

$$0 \otimes s_0 = 0 .$$

We understand S as the set of coordinates in a tensor.

Definition 2. A tensor with coordinates in S of order $d \in \mathbb{N}$ and leg dimensions $m_{[d]} = (m_0, \dots, m_{d-1})$, where $m_k \in \mathbb{N}$ for $k \in [d]$, is a map

$$\tau : \bigtimes_{k \in [d]} [m_k] \rightarrow S .$$

A tensor network with coordinates in S is a hypergraph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, where each node is decorated by a dimension m_v and each edge by a tensor of order $|e|$ and leg dimensions m_e .

2 Contractions

Definition 3. Given a semiring $(S, \oplus, \otimes, 0, 1)$ and a tensor network $\{\tau^e [X_e] : e \in \mathcal{E}\}$ with coordinates in S , the contraction of the tensor network keeping the variables to a subset $\mathcal{U} \subset \mathcal{V}$ open is the tensor

$$\langle \{\tau^e [X_e] : e \in \mathcal{E}\} \rangle_{[X_{\mathcal{U}}]}^{(S, \oplus, \otimes, 0, 1)}$$

with coordinates at indices $x_{\mathcal{U}}$ by

$$\langle \{\tau^e [X_e] : e \in \mathcal{E}\} \rangle_{[X_{\mathcal{U}}]}^{(S, \oplus, \otimes, 0, 1)} = \bigoplus_{x_{\mathcal{V} \setminus \mathcal{U}}} \bigotimes_{e \in \mathcal{E}} \tau^e [X_e = x_e] .$$

¹That is associativity, commutativity of $\oplus$ and identity of $s \oplus 0 = s$ for all $s \in S$.

The distributive property (iii) of the semiring ensures, that we can compute contractions by sub-contractions.

Theorem 1. *Given a disjoint partition of $\mathcal{E}$ into $\mathcal{E}^0, \mathcal{E}^1$, and a subset $\mathcal{U} \subset \mathcal{V}$, we define the sets $\mathcal{V}^0$ and $\mathcal{V}^1$ as the nodes appearing in $\mathcal{E}^0$, respectively $\mathcal{E}^1$. We then have*

$$\langle \{\tau^e[X_e] : e \in \mathcal{E}\} \rangle_{[X_{\mathcal{U}}]}^{(S, \oplus, \otimes, 0, 1)} = \langle \{\tau^e[X_e] : e \in \mathcal{E}^0\} \rangle_{[X_{\mathcal{V}^0 \cap (\mathcal{U} \cup \mathcal{V}^1)}]}^{(S, \oplus, \otimes, 0, 1)}, \langle \{\tau^e[X_e] : e \in \mathcal{E}^1\} \rangle_{[X_{\mathcal{V}^1 \cap (\mathcal{U} \cup \mathcal{V}^0)}]}^{(S, \oplus, \otimes, 0, 1)} \rangle_{[X_{\mathcal{U}}]}^{(S, \oplus, \otimes, 0, 1)}.$$

Proof. We use the associativity and commutativity of $\otimes$ to get for any index $x_{\mathcal{V}}$

$$\bigotimes_{e \in \mathcal{E}} \tau^e[X_e = x_e] = \left(\bigotimes_{e \in \mathcal{E}^0} \tau^e[X_e = x_e] \right) \otimes \left(\bigotimes_{e \in \mathcal{E}^1} \tau^e[X_e = x_e] \right).$$

Let us now keep $x_{\mathcal{U}}$ fixed and investigate the $\oplus$ operation over the indices $x_{\mathcal{V} \setminus \mathcal{U}}$. Distributivity of $\otimes$ over $\oplus$ implies

$$\sum_{x_{\mathcal{V} \setminus \mathcal{U}}} \bigotimes_{e \in \mathcal{E}} \tau^e[X_e = x_e] = \sum_{x_{(\mathcal{V}^0 \cap \mathcal{V}^1) \setminus \mathcal{U}}} \left(\sum_{x_{\mathcal{V}^0 \setminus (\mathcal{U} \cup \mathcal{V}^1)}} \bigotimes_{e \in \mathcal{E}^0} \tau^e[X_e = x_e] \right) \otimes \left(\sum_{x_{\mathcal{V}^1 \setminus (\mathcal{U} \cup \mathcal{V}^0)}} \bigotimes_{e \in \mathcal{E}^1} \tau^e[X_e = x_e] \right)$$

which is equivalent to

$$\langle \{\tau^e[X_e] : e \in \mathcal{E}\} \rangle_{[X_{\mathcal{U}}=x_{\mathcal{U}}]}^{(S, \oplus, \otimes, 0, 1)} = \langle \{\tau^e[X_e] : e \in \mathcal{E}^0\} \rangle_{[X_{\mathcal{V}^0 \cap (\mathcal{U} \cup \mathcal{V}^1)}]}^{(S, \oplus, \otimes, 0, 1)}, \langle \{\tau^e[X_e] : e \in \mathcal{E}^1\} \rangle_{[X_{\mathcal{V}^1 \cap (\mathcal{U} \cup \mathcal{V}^0)}]}^{(S, \oplus, \otimes, 0, 1)} \rangle_{[X_{\mathcal{U}}=x_{\mathcal{U}}]}^{(S, \oplus, \otimes, 0, 1)}.$$

Since therefore both tensors of the claimed contraction equation coincide on all coordinates, they are equal. $\square$

As a consequence, message passing schemes such as variable elimination generalize to semiring tensor network contractions.

3 Examples

3.1 Standard TN

Standard TN use the semiring

$$(\mathbb{R}, +, \cdot, 0, 1).$$

3.2 Constraint Satisfaction

We take $S = \{0, 1\}$ and interpret 0 as False, 1 as True. We define the disjunction and conjunction operations by

$$\vee[X_0, X_1] = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix} [X_0, X_1]$$

and

$$\wedge[X_0, X_1] = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} [X_0, X_1].$$

Tensors over $\{0, 1\}$ are interpreted as constraints, and their tensor networks as constraint networks.

The satisfiability of a constraint network $\{\tau^e[X_e] : e \in \mathcal{E}\}$ is the contraction

$$\bigvee_{x_{\mathcal{V}}} \bigwedge_{e \in \mathcal{E}} \tau^e[X_e = x_e] = \langle \{\tau^e[X_e] : e \in \mathcal{E}\} \rangle_{[\emptyset]}^{(\{0, 1\}, \vee, \wedge, 0, 1)}.$$

As an additional property, we have invariance of adding subcontractions to the network. In general tensor networks, this holds only for the support of subcontractions (being the operator mapping 0 to 0 and any other $s \in S$ to 1).

3.3 Maximum calculus

The maximal coordinate of a standard TN is the contraction wrt the semiring

$$(\mathbb{R}, \max, \cdot, 0, 1).$$

Analogously the minimal coordinate by

$$(\mathbb{R}, \min, \cdot, 0, 1).$$

These schemes only work, when taking the minimal coordiante keeping all variables open. When searching for the maximal/minimal coordiante of a contraction with some variables closed, we need to first coarse grain the tensor network.

These contractions are used in encoding-decoding schemes Mézard (2009); MacKay (2003). Message passing schemes involving the maximum semiring are known as max-product algorithms Koller and Friedman (2009).

3.4 Tropical semiring

The tropical semiring Liu et al. (2021) is $(\mathbb{R} \cup \{\infty\}, \min, +, \infty, 0)$.

References

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